



Scan for More...

- ✓ → Single Tick
- ✓✓ → Double Tick
- ✓✓ → Double Blue Tick

We use WhatsApp to send messages in the form of text, pictures or videos to our friends.

After we send a message, we see one of the three kinds of ticks near it. What do these ticks mean? How do they work?

Let's explore!



You might have heard the following statements.

If **you are ill**,
then **take rest**.



If **you are hungry**,
then **have a biscuit**.



If **you behave well**,
then **you can play video games**.



Did you notice the following?

- They have two parts: one **blue**, one **green**.
- The first part starts with IF, and the second part starts with THEN.
- The second part happens only if the first part happens.

We can say that the blue part is the **condition**, and the green part is the **result**.

Combined, they are known as **conditionals**.

A conditional is a statement where the result depends whether a condition is fulfilled.



Complete the following conditional statements:



Condition	Result
If you eat junk food,	then_____.
If you study well,	then_____.
If _____,	then you go to the next round.
If _____,	then you win the gold medal.

Let's dive deep into conditionals with more examples!

Look at these yummy chocolates.

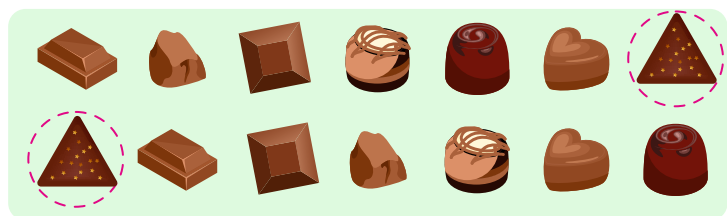
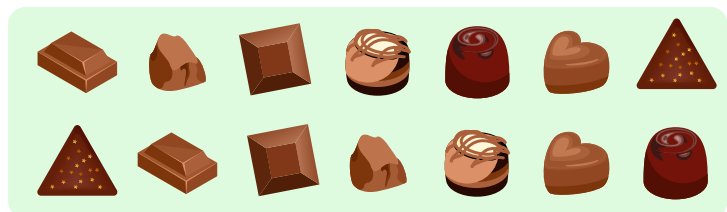
You are given this instruction:

If **the chocolate is a triangle**, then **you can eat it**.

Which chocolates will you eat?

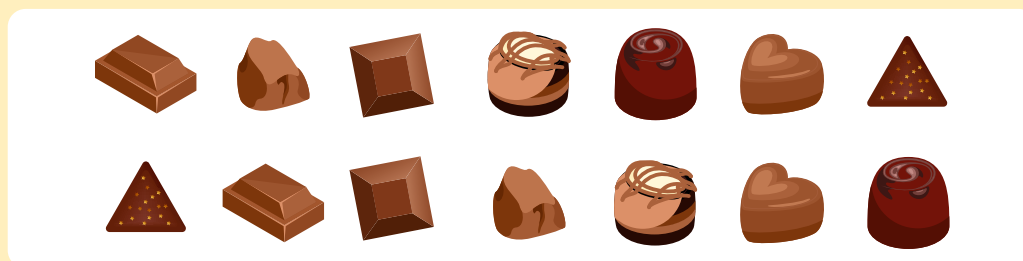
Even if you want to eat them all, you will eat just these two! This is because the condition is: "The chocolate is a triangle."

The others are not triangles, so you will not eat them.



In the same case, the condition is given as follows:

"If **the chocolate is not a square**, then **you can eat it**."

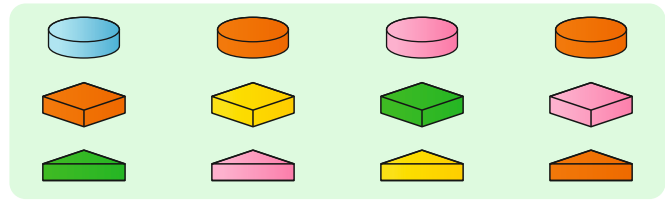


Write down how many chocolates you can eat: _____

Here are some candies.

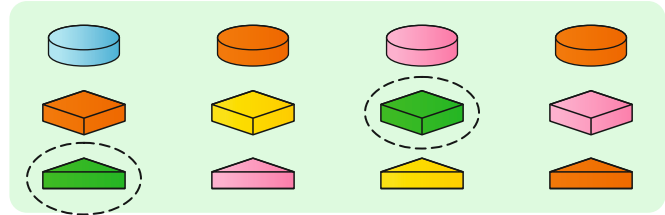
The instruction is: "If the candy is green, then you can eat it!"

Which candies will you eat?



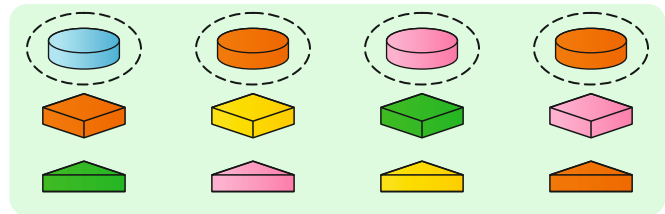
You will eat just these two, because the condition is: "The candy is green."

The other candies are not green.



In the same case as above, a kid ate these candies, following a different instruction.

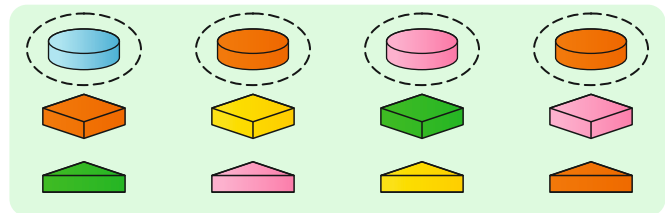
Which instruction could it be?



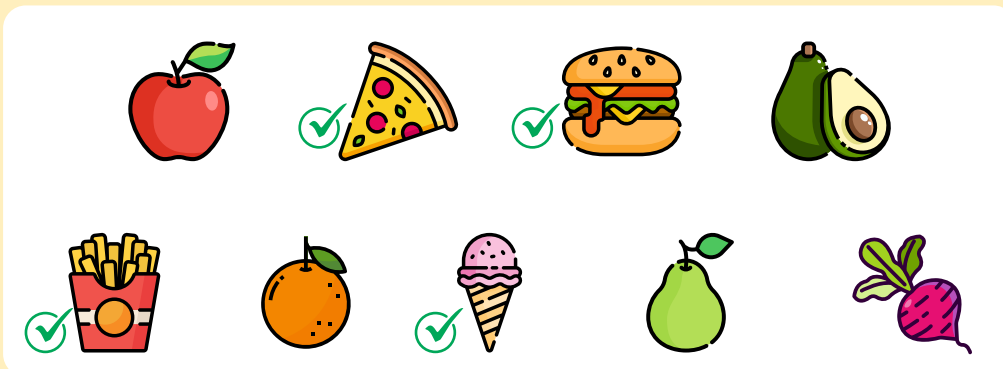
Here, the candies he ate were of different colors.

However, they all had the same round shape!

The instruction was: "If the candy is round, then you can eat it!"



A kid ticked these foods from the group.



What instruction did he follow?

"If the food is _____, then tick it!"

So far, we had only 1 condition. Sometimes, there are two conditions.

An example is given below:

If **you're a boy and you're 10 years old**,
then **you can join the boys football team**.

Here, to join the boys football team,
there are two conditions:

1. You're a boy.
2. You're 10 years old.



Both are joined by AND. It means that both the conditions need to be fulfilled.

- If you're a boy and **you're 9 years old**, then you CANNOT join the boys football team.
- If **you're a girl** and you're 10 years old, then you CANNOT join the boys football team.

Let's look at another example.

If **your marks are highest and you're good at a sport**, then **you can be the monitor**.

Here, BOTH these conditions need to be fulfilled:

1. Your marks are highest.
2. You're good at a sport.



Look at the following conditional:

“If **you're a girl and your favorite subject is Math**, then **join group A**.”

Which of these students can join group A?



Name	Aryan	Arush	Aditi	Krithi	Ramya	Sonia	Udbhav
Favorite subject	English	Maths	Maths	EVS	Maths	Maths	English
Gender	Boy	Boy	Girl	Girl	Girl	Girl	Boy
Group A (Yes / No)							

Sometimes, there are two conditions but only one of them or both need to happen for the result.

An example is given below:

"If you have a brother or a sister, then raise your hand."

Here, to raise the hand, there are two conditions:

1. You have a brother.
2. You have a sister.

The conditions are joined by OR. It means that only one or both must happen, as shown here:

If you have a brother, then raise your hand.



If you have a sister, then raise your hand.



If you have a brother and a sister, then raise your hand.



If you don't have brothers or sisters, then DON'T raise your hand.



Let's look at another example.

"If you like reading books OR playing chess, then raise your hand."

Here, to raise the hand, there are two conditions:

1. You like reading books.
2. You like playing chess.

Again, it means only one or both must happen, as shown here:

If you like reading books
and playing chess, then
raise your hand.



If you like playing chess but
don't like reading books,
then raise your hand.



If you like reading books but
don't like playing chess,
then raise your hand.

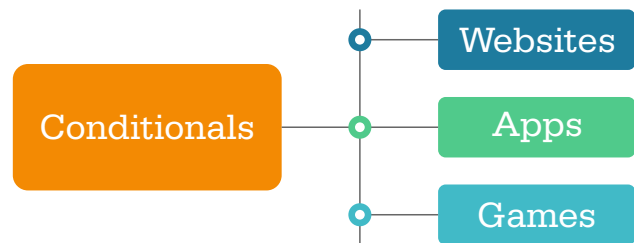


If you don't like reading books
and don't like playing chess,
then DON'T raise your hand.



Tick them according to this instruction:
 "If it is a fruit or a vegetable, then **tick it!**"

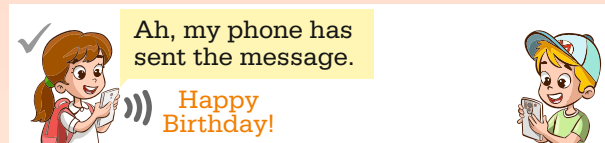
Conditionals are used heavily in computer programs, to make decisions. Especially, they are used here:



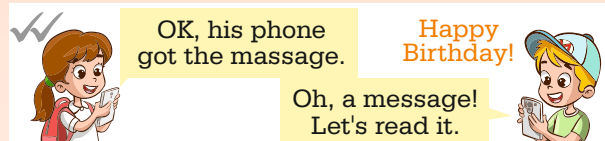
Now we understand conditionals, and their uses. Let's learn about the gray and blue ticks on WhatsApp!

WhatsApp programming uses conditionals. They are as follows:

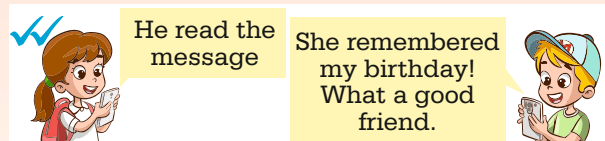
1. "If **the message has been sent**, then **show one gray tick**."



2. "If **the message has been received**, then **show two gray ticks**."



3. "If **the message has been read**, then **show two blue ticks**."



Thus, the ticks tell us the status of the message we sent using the conditionals!



LET'S RECAP!

- A conditional is a statement where the result depends on whether a condition is fulfilled.
- A conditional has two parts, the condition and the result.
- Sometimes, a conditional can have two conditions, joined together using 'and' or 'or'.
- When two conditions are joined using 'and', it means that both of them need to happen for the result to happen.
- When two conditions are joined using 'or', it means that only one of them is enough for the result to happen.
- In computer programs, conditionals are used to make decisions.



A. MULTIPLE CHOICE QUESTIONS

1. The teacher said this to the students:

"If your T-shirt number is even, **come forward!**"

How many students came forward?



(a) 2

(b) 3

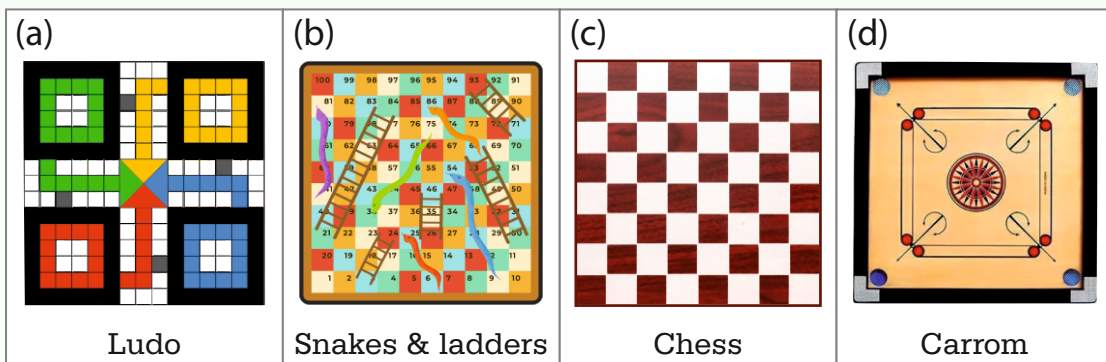
(c) 4

(d) 5

2. Look at the conditional below:

"If the game is only a 2-player game, then **buy it!**"

Which board game should you buy?



3. Look at this conditional:

"If the food is sweet and cold, then **buy it!**"

Which foods will you buy?



4. From a group of animals, some were ticked.

					
Lion	Monkey	Squirrel	Cat	Dog	Rat

Under which condition were the animals ticked?

- (a) If the animal is big and has four legs, then tick it.
- (b) If the animal is brown and big, then tick it.
- (c) If the animal climbs trees and is brown, then tick it.
- (d) If the animal is big and climbs trees, then tick it.

B. FILL IN THE BLANKS

1. _____ is a statement where the result depends whether a condition is fulfilled.
2. Conditions joined using _____ means both of them are needed for the result.
3. Conditions joined using _____ means one or both of them are enough for the result.

C. TRUE OR FALSE

1. A conditional is a statement with only one condition and a result.
2. When two conditions are joined using 'and', it means only one of them occurring is enough for the result to happen.
3. In computer programs, conditionals are used to make decisions.
4. Two conditions joined by 'or' means both need to occur.

D. VERY SHORT ANSWER - TYPE QUESTIONS

1. What is a statement in which the result depends on whether a condition is fulfilled or not?

E. SHORT ANSWER - TYPE QUESTIONS

1. Write examples of two real-life conditional statements.

2. What is the difference between two conditions combined using AND and OR?

ACTIVITY TIME!

1. This is a group activity.
 - a. All the students are given two blank cards each.
 - b. Each student writes down a condition on one card, and the result on the other.
 - c. The result cards are mixed up in a pile.
 - d. Then each student is randomly given one card from the pile.
 - e. The aim here is to find the correct partner of the 'Condition' card they have, by talking to other students and exchanging their cards.
 - f. When a student has found his correct pair, he goes to the front of the class, and tells everyone the conditional he made.
 - g. This continues till everyone has found their conditional pair.
2. In turn-based games, how much you advance is based on your choices.

Try them at this link :

<https://www.crazygames.com/t/turn-based>

What did I learn?

- ☐ I can explain what conditionals are with the help of examples
- ☐ I can demonstrate how a conditional can have two conditions
- ☐ I can explain usage of conditionals in programming